

Questions, then easy solutions

Twenty-five questions across all four foundation topics

YOUR AIM TODAY

Complete the question section without notes, then use the back section to understand and correct every answer.

YOUR RULE

Keep the solution section closed until every question has an attempt. Show working. Mark in another colour and write the first wrong step.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

INSTRUCTIONS

How to complete this booklet

Suggested time: 90 minutes questions, then 45 minutes marking.

Stage	Action
Questions	Complete Sections A-E without notes.
Mark	Use the easy solutions only after all questions are attempted.
Correct	Circle the first wrong line and state the correct rule.
Retest	Redo every incorrect question from a blank page tomorrow.

TARGET

22-25 correct: secure. 18-21: repair specific gaps. 14-17: reteach two weakest topics. Below 14: return to the notes booklet before retesting.

Quadratics

Show every important step.

1. Expand and simplify $(x-5)(2x+3)$.

2. Complete the square: x^2+6x-7 .

3. Hence state the turning point of $y=x^2+6x-7$.

4. Solve $x^2+6x-7=0$.

5. Solve $2x^2+x-6=0$.

Discriminants and inequalities

Show every important step.

1. Find the discriminant of $x^2-4x+8=0$ and state the number of real roots.

2. Find k if $x^2+10x+k=0$ has a repeated root.

3. Find the values of p for which $x^2+px+16=0$ has no real roots.

4. Solve $x^2-5x+6<0$.

5. Solve $2x^2-x-3\geq 0$.

Functions

Show every important step.

GIVEN

Let $f(x)=3x-2$ and $g(x)=x^2+4$.

1. Find $f(-2)$.

2. Find $f(a+1)$.

3. Find $fg(x)$.

4. Find $gf(x)$.

5. Find $f^{-1}(x)$, then explain why g has no inverse on all real numbers.

Coordinate geometry

Show every important step.

GIVEN

Points $A=(-2,1)$ and $B=(4,9)$.

1. Find the gradient of AB.

2. Find the midpoint of AB.

3. Find the exact length AB.

4. Find the equation of AB.

5. Find the perpendicular bisector of AB.

Circles

Show every important step.

GIVEN

Circle C: $x^2 + y^2 - 4x + 6y - 12 = 0$.

1. Write C in completed-square form; state its centre and radius.

2. Show that $P=(5,1)$ lies on C.

3. Find the tangent to C at P.

4. Find the intersections of C with $y=-3$.

5. Write the equation of a circle with centre $(-1,2)$ and radius 3.

Open only after completing Sections A-E

Read one line, compare one line, then correct without copying.

EASY EXPLANATION RULE

Each solution names the essential move. If your first move was different, write the missing rule before reading further.

My score before corrections: ____ / 25

Quadratics

Q	Easy working and answer
1	1. Multiply all four pairs: $2x^2+3x-10x-15=2x^2-7x-15$.
2	2. Half of 6 is 3: $x^2+6x-7=(x+3)^2-9-7=(x+3)^2-16$.
3	3. The minimum square occurs at $x=-3$, and $y=-16$. Turning point $(-3,-16)$.
4	4. Use $(x+3)^2-16=0$. Then $(x+3)^2=16$, $x+3=\pm 4$, so $x=1$ or $x=-7$.
5	5. Factor: $2x^2+x-6=(2x-3)(x+2)$. Therefore $x=3/2$ or $x=-2$.

CORRECTION TASK

Close the solutions and redo every incorrect question from the beginning. Do not copy the model.

Discriminants and inequalities

Q	Easy working and answer
1	1. $a=1, b=-4, c=8$. $D=(-4)^2-4(1)(8)=16-32=-16$. Since $D<0$, there are no real roots.
2	2. Repeated means $D=0$: $10^2-4k=0$, so $100-4k=0$ and $k=25$.
3	3. No real roots means $p^2-64<0$. Therefore $p^2<64$ and $-8<p<8$.
4	4. $(x-2)(x-3)<0$. The upward parabola is negative between roots: $2<x<3$.
5	5. $(x+1)(2x-3)\geq 0$. The upward parabola is non-negative outside: $x\leq -1$ or $x\geq 3/2$.

CORRECTION TASK

Close the solutions and redo every incorrect question from the beginning. Do not copy the model.

Functions

Q	Easy working and answer
1	1. $f(-2)=3(-2)-2=-6-2=-8$.
2	2. $f(a+1)=3(a+1)-2=3a+3-2=3a+1$.
3	3. $fg(x)=f(x^2+4)=3(x^2+4)-2=3x^2+10$.
4	4. $gf(x)=g(3x-2)=(3x-2)^2+4=9x^2-12x+8$.
5	5. $y=3x-2$; swap: $x=3y-2$; $y=(x+2)/3$, so $f^{-1}(x)=(x+2)/3$. g is not one-to-one because $g(t)=g(-t)$.

CORRECTION TASK

Close the solutions and redo every incorrect question from the beginning. Do not copy the model.

Coordinate geometry

Q	Easy working and answer
1	1. $m = (9-1)/(4-(-2)) = 8/6 = 4/3$.
2	2. Midpoint $= ((-2+4)/2, (1+9)/2) = (1, 5)$.
3	3. Changes are 6 and 8: distance $= \sqrt{6^2 + 8^2} = 10$.
4	4. Through A with $m = 4/3$: $y - 1 = (4/3)(x + 2)$, so $y = (4/3)x + 11/3$.
5	5. Perpendicular $m = -3/4$ through midpoint $(1, 5)$: $y - 5 = (-3/4)(x - 1)$.

CORRECTION TASK

Close the solutions and redo every incorrect question from the beginning. Do not copy the model.

Circles

Q	Easy working and answer
1	1. $(x-2)^2 + (y+3)^2 = 25$. Centre (2, -3), radius 5.
2	2. At P: $(5-2)^2 + (1+3)^2 = 3^2 + 4^2 = 25$, so P lies on C.
3	3. Radius gradient $= (1 - (-3)) / (5 - 2) = 4/3$, so tangent $m = -3/4$. $y - 1 = (-3/4)(x - 5)$, or $3x + 4y = 19$.
4	4. Set $y = -3$: $(x-2)^2 = 25$, so $x = 7$ or -3 . Points (7, -3) and (-3, -3).
5	5. Centre (-1, 2), radius 3: $(x+1)^2 + (y-2)^2 = 9$.

CORRECTION TASK

Close the solutions and redo every incorrect question from the beginning. Do not copy the model.

What must change next time

The first wrong step is the useful information.

Q	First wrong step	Error type	Correct rule

Three rules I can now explain without notes:

SEND FOR REVIEW

Send the completed question pages and correction page. Ask for three follow-up questions from the weakest topic.